

FINAL PROJECT REPORT

Store-Level Demand Forecasting and Inventory Optimization for the Rossmann Retail Chain

Sector: Retail Analytics / Supply Chain Optimization

Team: Raghav gupta,Aaryan yadav,Vaageesh Kumar Singh,Kumar sanu,Balanagu sesha srikar

Institute: Newton School Of Technology

1. Executive Summary

Retail chains operate in a highly dynamic environment where customer demand fluctuates due to seasonality, promotions, holidays, and store-level variations. Inaccurate forecasting often results in stockouts, lost revenue, and excessive inventory holding costs.

This project analyzes historical sales data from the Rossmann retail chain to develop store-level demand forecasting insights and provide inventory optimization recommendations.

Using historical daily sales data, promotional flags, holiday indicators, and store characteristics, we performed:

- Data cleaning and preparation in Google Sheets
- Exploratory data analysis
- Trend and seasonal analysis
- Forecast modeling for future demand
- Inventory optimization simulation

Key insights reveal that promotions significantly increase sales, seasonal patterns are consistent, and store types differ in revenue generation. Based on forecasted demand, we propose reorder strategies and safety stock adjustments to reduce stockout risk while minimizing overstock.

The final dashboard supports data-driven inventory decisions at the store level.

2. Sector & Business Context

Retail Sector Overview

The global retail industry faces increasing complexity due to:

- Multi-store operations
- Dynamic pricing and promotions
- Seasonal buying patterns
- Competitive pressure

Drugstore chains like Rossmann operate thousands of outlets, making centralized forecasting essential.

Current Challenges

- Demand variability
- Promotion-driven spikes
- Holiday-based fluctuations
- Inventory misalignment
- Operational inefficiencies

Why This Problem Was Chosen

Demand forecasting directly impacts:

- Revenue
- Customer satisfaction
- Inventory holding costs
- Working capital efficiency

Improving forecast accuracy improves profitability.

3. Problem Statement & Objectives

Problem Statement

Retail stores experience fluctuating demand due to seasonality, promotions, and holidays. Without accurate forecasting, stores face:

- Stockouts (lost sales)
- Excess inventory (high holding cost)
- Poor replenishment planning

Objectives

1. Analyze historical store-level sales patterns
2. Identify drivers of demand (promo, holiday, seasonality)
3. Develop demand forecasting insights
4. Provide inventory optimization recommendations
5. Build a dashboard for operational decision-making

Success Criteria

- Clear seasonal patterns identified
 - Promotion impact quantified
 - Forecast generated for next 30 days
 - Inventory recommendations supported by logic
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4. Data Description

Dataset Source

Dataset obtained from:- **Kaggle**

Competition:- **Rossmann Store Sales**

Data Structure

Primary Dataset: [train.csv](#)

Key Columns

- Store – Store ID
- Date – Sales date
- Sales – Daily revenue
- Customers – Customer count
- Open – Store open indicator
- Promo – Promotion running (1/0)

- StateHoliday – Public holiday
- SchoolHoliday – School holiday

Data Size

- ~6,000+ rows
- Daily store-level records
- Multi-year time coverage

Data Limitations

- No product-level data
 - No inventory quantity data
 - External macroeconomic variables missing
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5. Data Cleaning & Preparation

(All cleaning executed in Google Sheets as per requirement.)

Steps Performed:

1 Missing Values

- Removed rows where Sales = 0 & Open = 0
- Treated blank holiday values as “No Holiday”

2 Outlier Treatment

- Identified extreme spikes using conditional formatting
- Verified whether linked to promotions before removal

3 Transformations

- Extracted:
 - Year
 - Month
 - Day
 - Week number
- Created new columns:
 - Avg Sales per Customer
 - Sales Growth %

4 Feature Engineering

- Promotion Impact Flag
- Seasonal Index
- Monthly Average Sales

5 Assumptions

- Sales = 0 when store closed
- Promotional impact uniform across product categories

6. KPI & Metric Framework

Sales Performance KPIs

1. Top Revenue Store Type: A
 2. Sales Lift During Promotion
 3. Average Daily Sales
 4. Highest Avg Sales Format: Type B - Extra
 5. Average Daily Customers
 6. Revenue Per Customer
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7. Exploratory Data Analysis (EDA)

Trend Analysis

- Sales show strong seasonality
- Year-end months show peak demand
- Weekend sales differ from weekdays

Promotion Analysis

- Sales increase significantly during promotions
- Customer count strongly correlates with revenue

Distribution Analysis

- Sales distribution is right-skewed
- Few high-performing stores drive major revenue

Correlation Analysis

Strong positive correlation between:

Customers & Sales

Moderate correlation between:

Promo & Sales

8. Advanced Analysis

Demand Forecasting

Using time-series trend extrapolation:

- Identified monthly seasonal patterns
- Applied moving average forecasting
- Generated 30-day sales projection

Scenario Analysis

Scenario 1: No Promotion

Scenario 2: With Promotion

Promotional periods show ~15–25% uplift.

Risk Analysis

Stockout risk highest during:

- Holiday weeks
 - Major promotional campaigns
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9. Dashboard Design

Dashboard Built in Google Sheets using:

- Pivot tables
- Slicers
- Interactive filters
- Forecast summary section

Dashboard Views

- Monthly Trend
- Store-Level Performance
- Promotion Impact
- Forecast Summary
- Inventory Simulation

Filters

- Store ID
 - Month
 - Year
 - Promo status
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10. Key Insights Summary

1. Promotions significantly increase sales.
 2. Customer count is the strongest predictor of revenue.
 3. Clear monthly seasonal patterns exist.
 4. Certain store types outperform consistently.
 5. Holiday periods create demand spikes.
 6. Inventory risk highest during peak season.
 7. Some stores show unstable demand patterns.
 8. Sales growth fluctuates across quarters.
 9. Over-reliance on promotions reduces baseline stability.
 10. Forecasting improves replenishment planning visibility.
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11. Recommendations

Recommendation 1: Optimize Promotion Timing

Mapped to: Promotion Impact Insight

Impact: Higher ROI per campaign

Recommendation 2: Dynamic Safety Stock

Mapped to: Seasonal Index

Impact: Reduce stockouts

Recommendation 3: Store-Specific Reorder Levels

Mapped to: Demand Variability

Impact: Lower holding cost

Recommendation 4: Holiday-Based Inventory Boost

Mapped to: Holiday Spike Insight

Impact: Prevent lost sales

12. Impact Estimation

If forecast accuracy improves by 10%:

- 5% reduction in stockouts
- 3–5% revenue increase
- 8–10% reduction in excess inventory
- Improved working capital efficiency

Estimated Financial Benefit:

Significant cost optimization at scale across stores.

13. Limitations

- No SKU-level inventory data
 - Forecast model basic (Google Sheets)
 - External economic factors excluded
 - Assumes uniform lead time
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14. Future Scope

- Implement ML-based forecasting (ARIMA / XGBoost)
 - Integrate supplier lead time data
 - Include pricing & competitor data
 - Real-time dashboard automation
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15. Conclusion

This project demonstrates how store-level demand forecasting can significantly improve inventory optimization in retail operations.

Through structured data cleaning, KPI analysis, forecasting, and dashboard visualization, actionable business insights were generated.

The framework developed enables data-driven decision-making, reducing operational inefficiencies and improving profitability for the Rossmann retail chain.

16. Appendix

Data Dictionary

Store – Unique store ID

Sales – Daily sales revenue

Customers – Number of customers

Promo – Promotion flag

StateHoliday – Public holiday

SchoolHoliday – School holiday

Tools Used

- Google Sheets

- Pivot Tables
- Excel Functions
- Time-Series Trend Analysis

17. Contribution Matrix (Mandatory)

Team-Member	Dataset & Sourcing	Cleaning	KPI & Analysis	Dashb oard	Report Writing	PPT	Overall Role
Balanag u sesha srikar	☒	☒					
Raghav Gupta	☒				☒		
Aaryan Yadav					☒	☒	
Vaagesh Singh			☒	☒			
Kumar Sanu		☒		☒			
shaik junaaid sami			☒			☒	